UNIVERSITI TEKNOLOGI MALAYSIA

SKEF 3003

ELECTRONICS

2025/2026 SEMESTER I

PROJECT LAB REPORT

Title of Experiment:	Construction, Measurement, and Analysis of Diode, LED, and Zener Diode Circuits (Team Hands-On Session)
Section	EB1
Date and Time of Experiment:	2025
Lecturer Name:	Ts.Chua Tien Han

Group Members	**Matric No.**
LUBNA AL HAANI BINTI RADZUAN	A23CS0107
NURAINI ISMA BINTI MAHADIR	A23SC0298
MUHAMMAD AMIRUL HAKIM BIN ROSMAINI	A23SC0176

INTRODUCTION

A diode is a semiconductor device that allows current to flow in one direction while restricting it in the opposite direction. Diodes are widely used in electronics for rectification,and voltage regulation.There are several types of diodes,each with specific characteristics and applications which are:

1. Normal (PN Junction) Diode

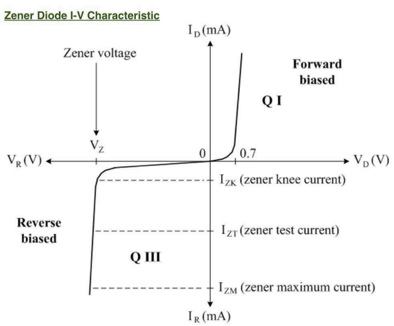
A PN junction diode is formed by joining p-type make contact with n-type semiconductor.It conducts current when forward-biased (positive voltage on the p-side) and blocks current when reverse biased until breakdown occurs. The forward voltage for silicon diodes is approximately 0.7 V.

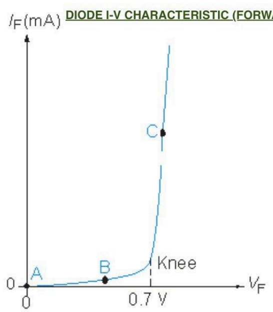
2. Light Emitting Diode (LED)

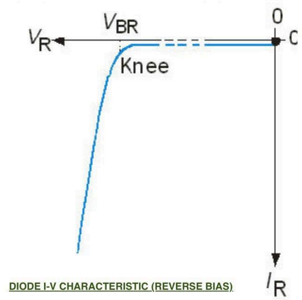
LEDs are diodes that emit light when there is forward current due to electron hole recombination. They typically require a higher forward voltage (1.8–3.3 V depending on color) and are not intended for reverse-bias operation. Their I–V curve is similar to a normal diode but shifted to a higher forward voltage.

3. Zener Diode

Zener diodes are designed to operate in reverse bias. They behave like a normal diode in forward bias (0.7 V conduction) but allow current to flow in reverse once the applied voltage reaches the Zener breakdown voltage.







In forward bias,all diodes allow current to flow once the voltage reaches a certain level (treshold),while in reverse bias normal diodes and LED block current but zener diode start conducting sharply when the reverse voltage reaches the zener voltage.

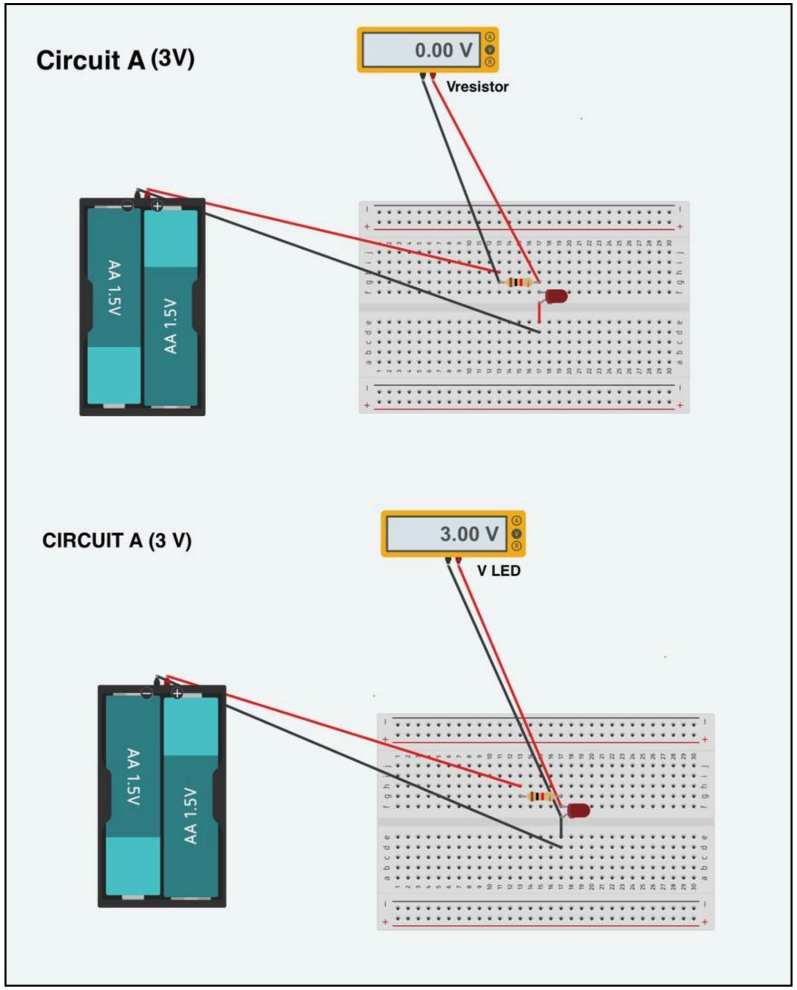
EXPERIMENT METHODOLOGY

In this experiment, we received the electronic kit and identified all the components listed which includes a breadboard, multimeter, resistor (1kΩ), diode, LED, Zener diode, battery holders, two 1.5V batteries and one 9V battery. The circuit was constructed using configuration (a) to (f) as per instructions. The circuit was assembled on the breadboard using the specified components. Once the connections were thoroughly checked, the power supply was then connected. The DC voltage source (Vs) was set to the specified value which is 3V and 9V for configuration (a) to (d) and 1.5V, 3V and 9V for configuration (e) and (f). The physical behavior of the circuit was observed for each voltage setting, paying additional attention as to whether the LED was ON or OFF and whether it was low intensity or high intensity. As per usual, a multimeter was used to measure and record the voltages for each circuit and voltage setting; voltage across resistor, Vresistor, voltage across the standard diode, Vdiode, voltage across Zener diode, Vzener. The biasing condition, whether forward bias or reverse was determined based on the measured voltages and the circuit configuration

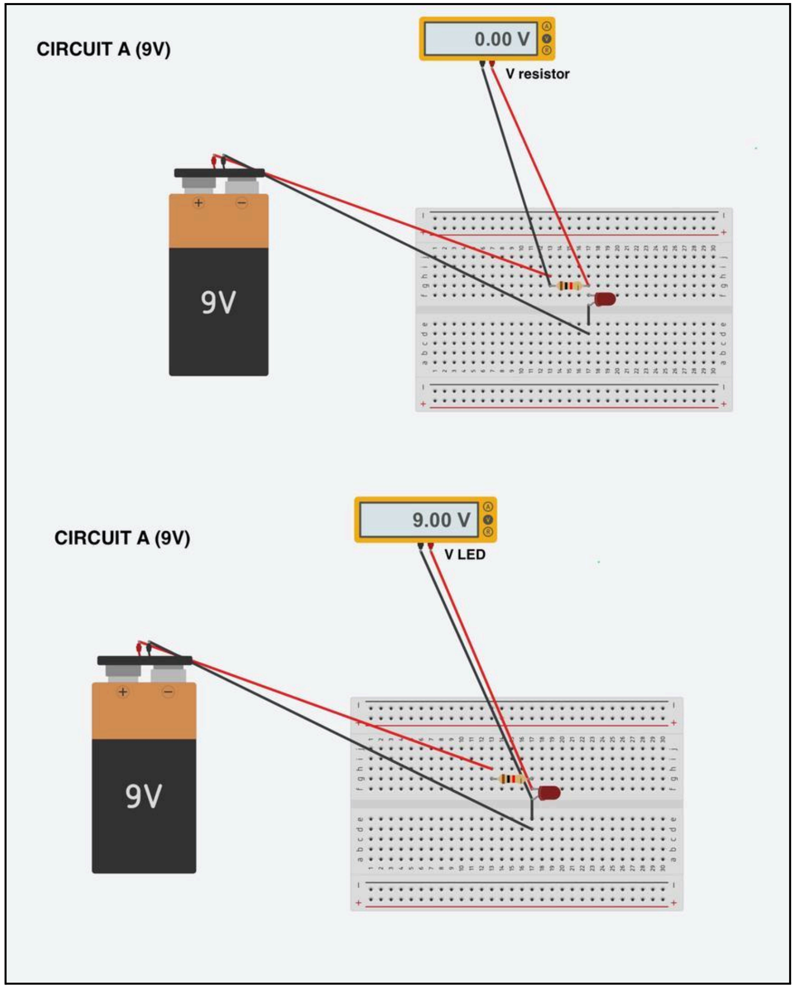
After the experiment was conducted, all the measurement data and observations noted was compiled into a single table. For each circuit configuration, a theoretical analysis was performed by using the practical diode model (0.7V for Si diodes, 2V for LED and Zener voltage for the zener diode in breakdown) to calculate the expected voltages and currents. Finally, a formal report was prepared, with comparison of the measured data with theoretical calculations, discussing the differences between theoretical and experimental values and summarizing the conclusions.

CIRCUIT DIAGRAM

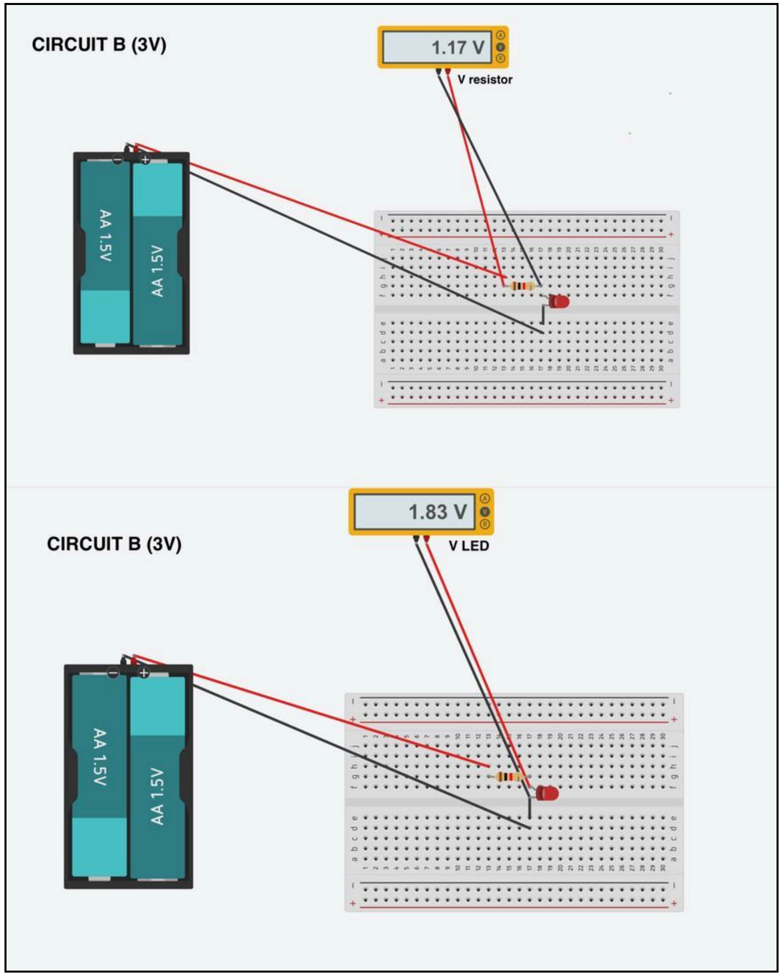
Circuit A (3V):



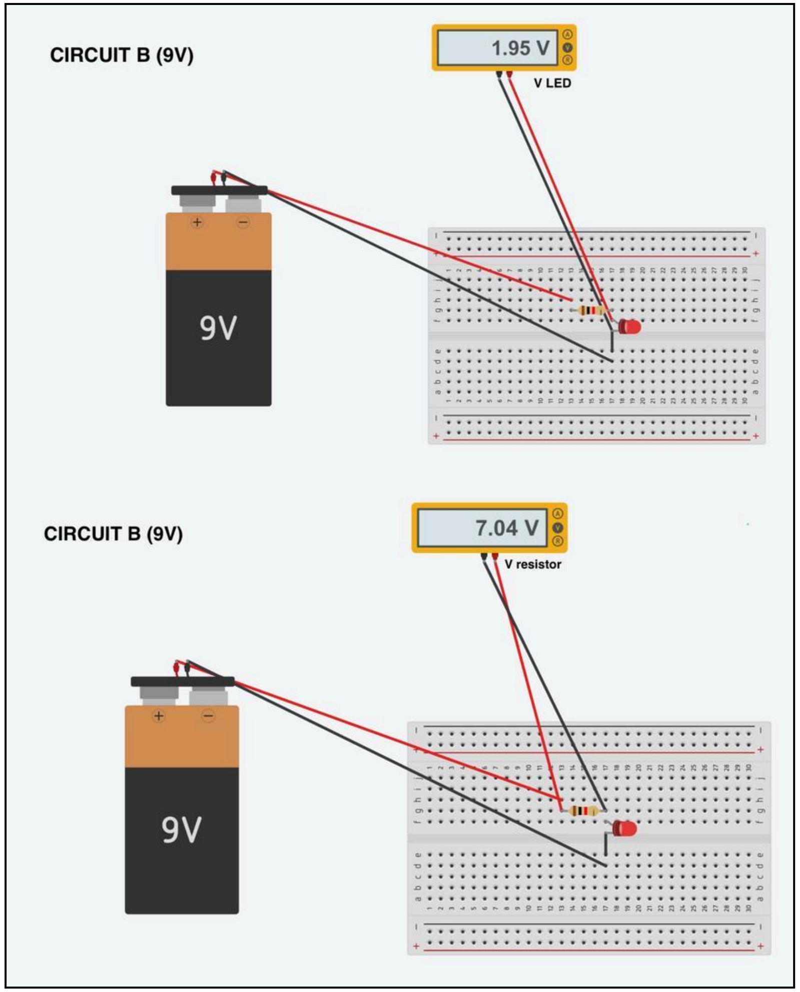
Circuit A (9V):



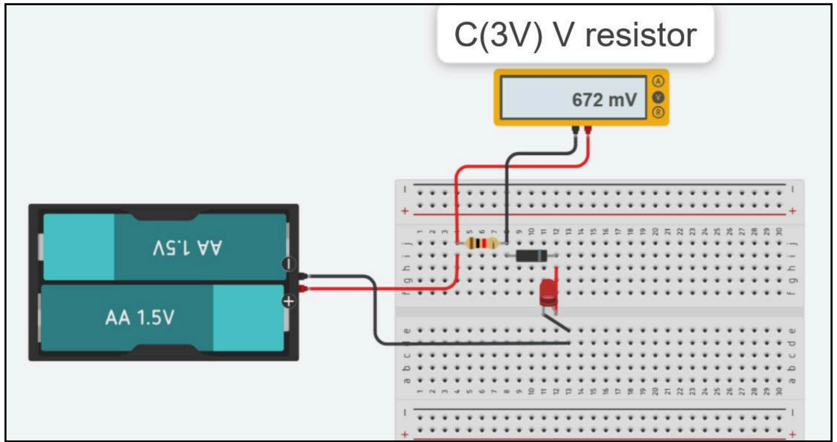
Circuit B (3V):

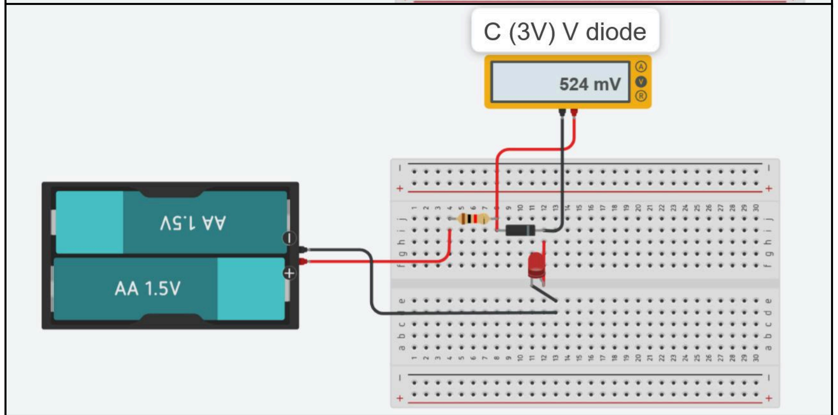


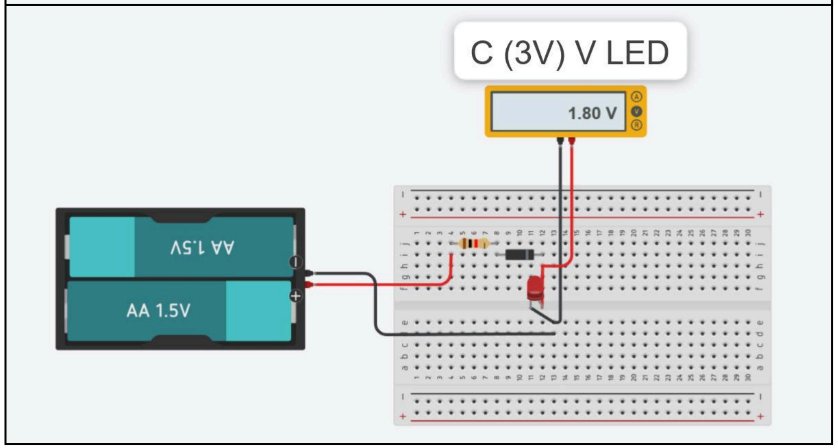
Circuit B (9V):



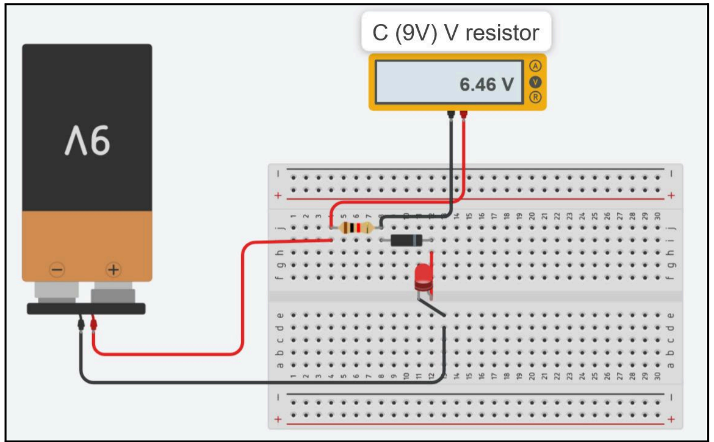
Circuit C (3V):

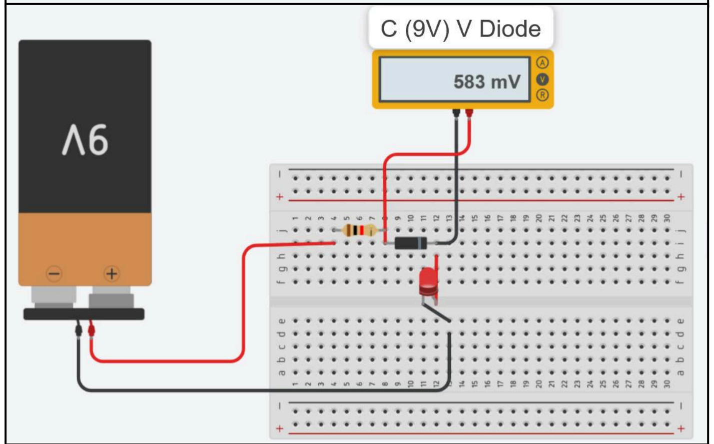


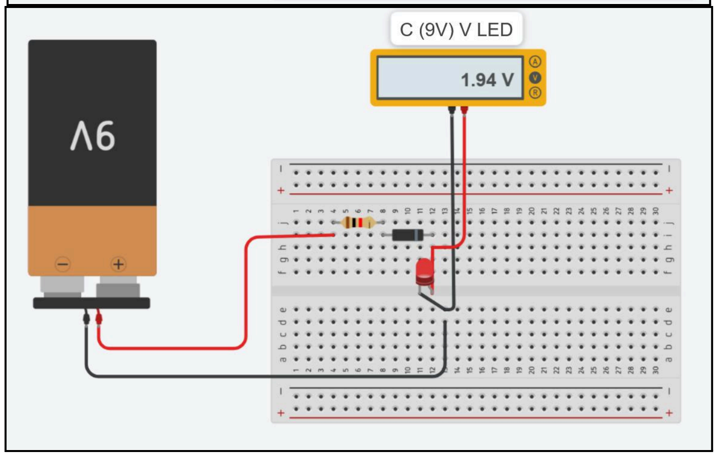




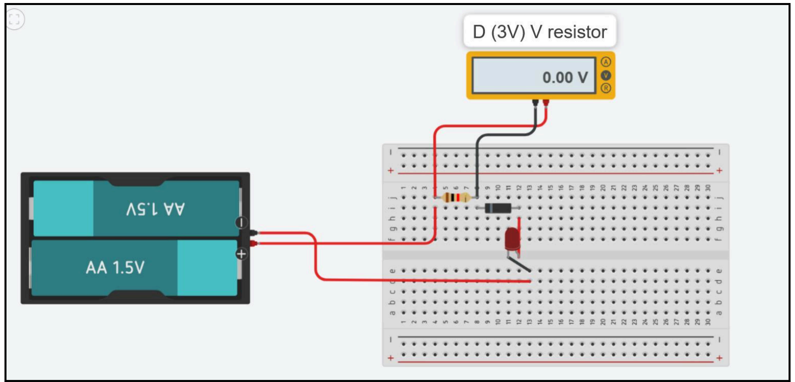
Circuit C (9V):

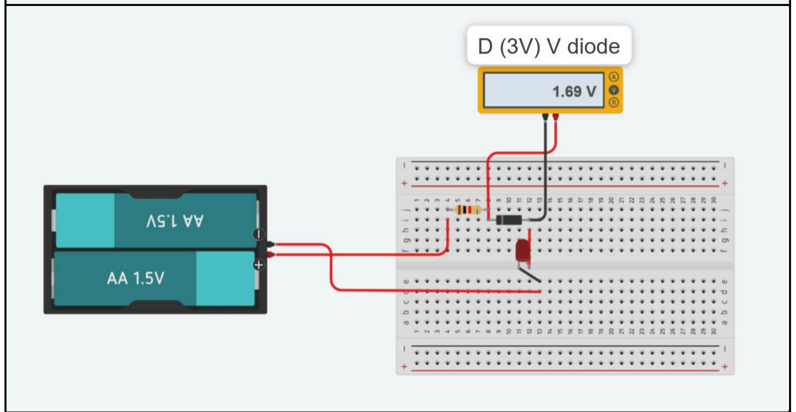


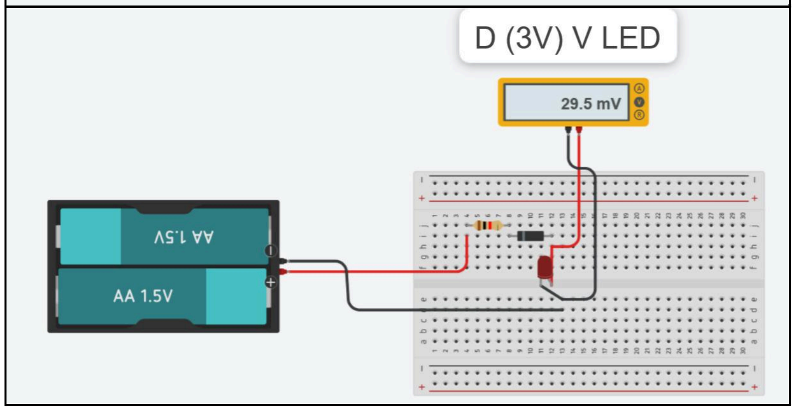




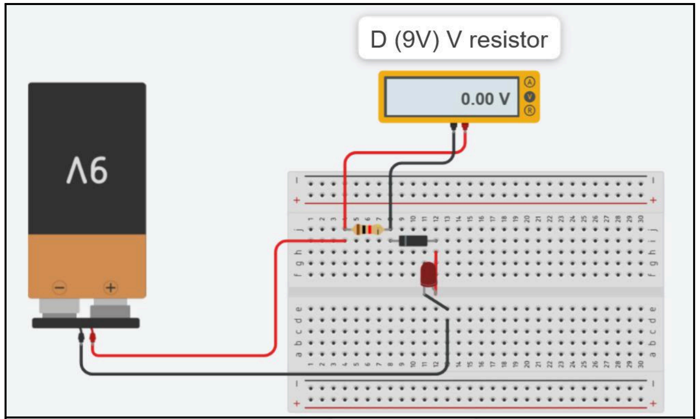
Circuit D (3V):

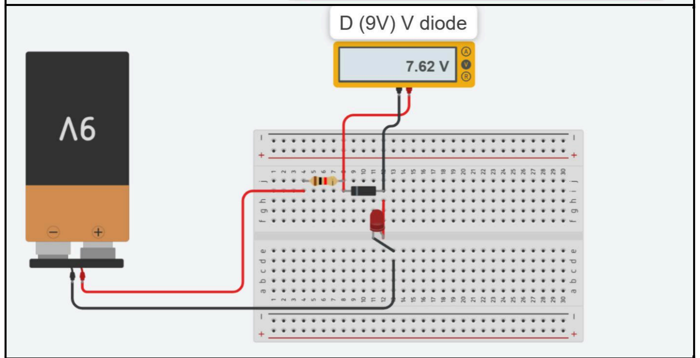


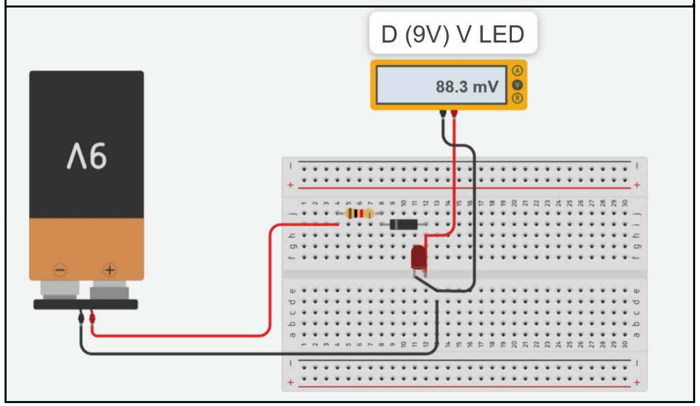




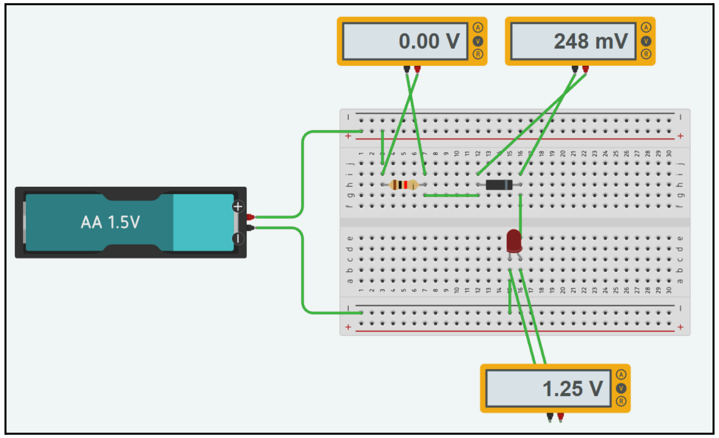
Circuit D (9V):



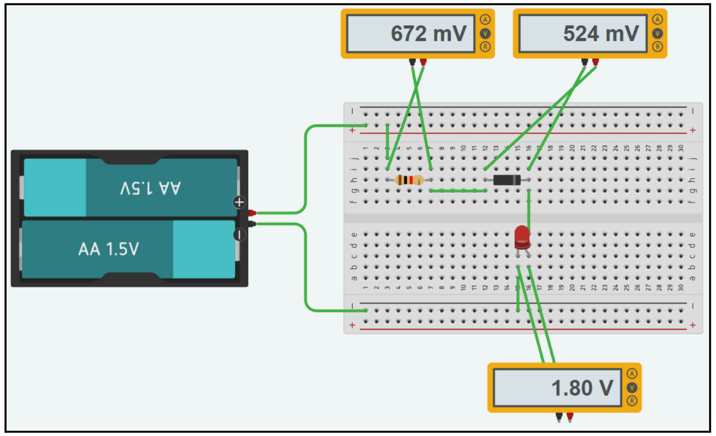




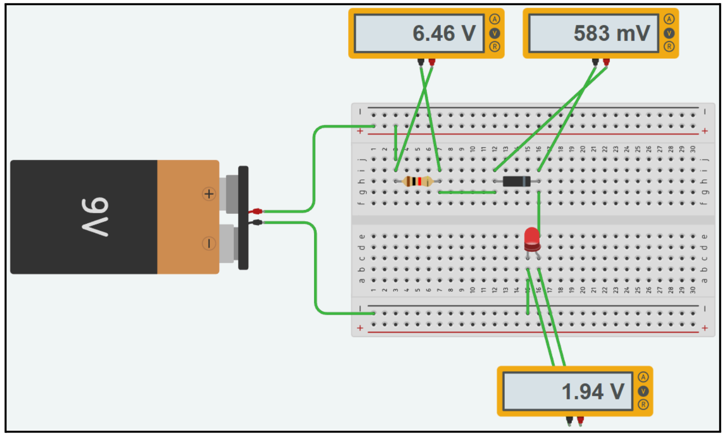
Circuit E (1.5V):



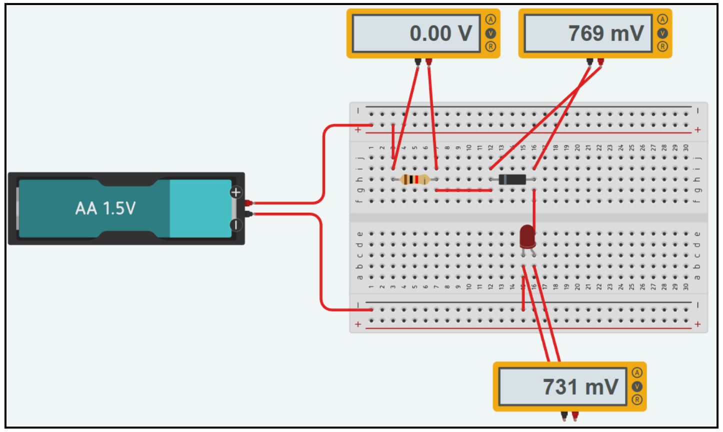
Circuit E (3V):



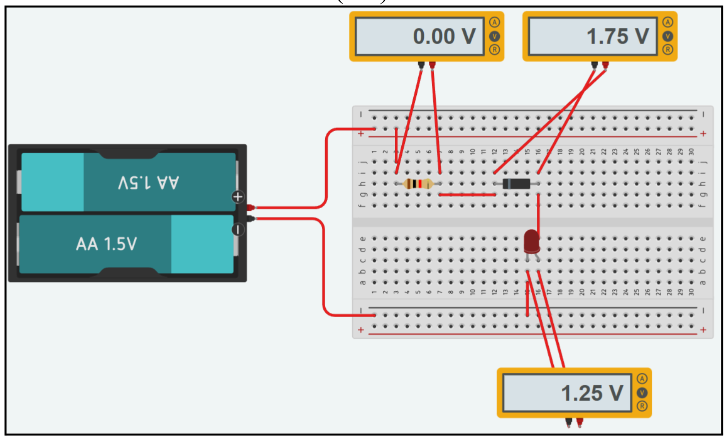
Circuit E (9V):



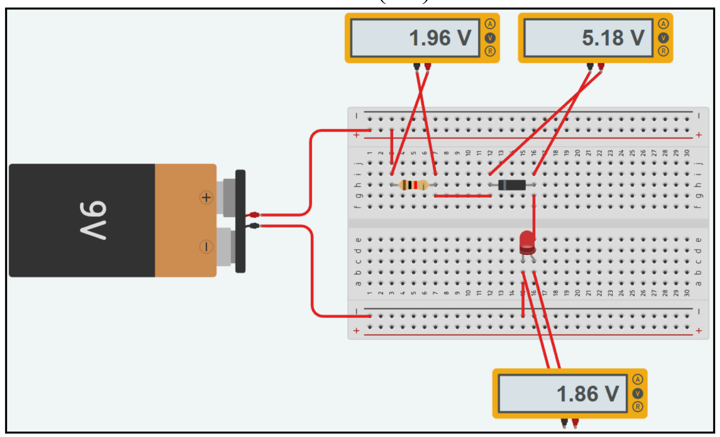
Circuit F (1.5V):



Circuit F (3V):



Circuit F (9V):



MEASUREMENT DATA & OBSERVATION

Experiment	DC SOURCE	RESISTOR	LED		DIODE		ZENER DIODE		Observation on the LED status (off/on,light intensity/high light intensity)
	V_s (V)	$V_{resistor}$	LED Biasing (FB/RB)	V_{LED} (V)	Diode Biasing (FB/RB)	V_{Diode} (V)	Biasing (FB/RB)	V_{Zener} (V)	
a	3	0.00	RB	3.00					OFF
	9	0.00	RB	9.00					ON, HIGH LIGHT INTENSITY
b	3	1.17	FB	1.83					ON, LOW LIGHT INTENSITY
	9	7.04	FB	1.95					ON, HIGH LIGHT INTENSITY
c	3	0.672	FB	1.80	FB	0.524			ON, LOW LIGHT INTENSITY
	9	6.46	FB	1.94	FB	0.583			ON, HIGH LIGHT INTENSITY
d	3	0.00	FB	0.0295	RB	1.69			OFF
	9	0.00	FB	0.0883	RB	7.62			ON, HIGH LIGHT INTENSITY
e	1.5	0.00	FB	1.25			FB	0.248	OFF
	3	0.672	FB	1.80			FB	0.524	ON, LOW LIGHT INTENSITY
	9	6.46	FB	1.94			FB	0.583	ON, HIGH LIGHT INTENSITY
f	1.5	0.00	FB	0.731			RB	0.769	OFF
	3	0.00	FB	1.25			RB	1.75	OFF
	9	1.96	FB	1.86			RB	5.18	ON, LOW LIGHT INTENSITY

THEORETICAL CIRCUIT ANALYSIS FOR EACH CONFIGURATION
Assume:

- Si Diode Forward Voltage (Vf diode) = **0.7 V**
- LED Forward Voltage (Vf LED) = **2.0 V**
- Zener Diode Forward Voltage = **0.7 V**
- Zener Diode Reverse Breakdown Voltage (Vz) = **5.0 V** (determined from Circuit (f) data)
- Resistor R = **1 kΩ** for all circuits

Circuit (a) : Vs → Resistor → LED (RB) → Ground

I = 0 A
Vr = I × R = 0 V
VLED = Vs
Vs for 3V source = 3V
Vs for 9V source = 9V

***Follows reverse-biased LED behaviour.**

Circuit (b): Vs → Resistor → LED (FB) → Ground

VLED = 2.0 V
Vr = Vs - 2.0V
I = Vr / 1000

Vr at Vs= 3V, Vr = 3 V - 2.0 V = 1.0V
I = Vr / 1000 = 1.0V/1000 = 1.0 mA

Vr at Vs= 9V, Vr = 9 V - 2.0 V = 7.0V
I = Vr / 1000 = 7.0V/1000 = 7.0 mA

***Follows forward-biased LED behaviour.**

Circuit (c): Vs → Resistor → Diode (FB) → LED (FB) → Ground

Vr = Vs - 0.7 - 2.0 = Vs - 2.7
I = Vr / 1000

Vr at Vs= 3V, Vr = 3 V - 2.7 V = 0.3V
I = Vr / 1000 = 0.3V/1000 = 0.3 mA

Vr at Vs= 9V, Vr = 9 V - 2.7 V = 6.3V
I = Vr / 1000 = 6.3V/1000 = 6.3 mA

***Vdiode is lower at lower currents.**

Circuit (d): Vs → Resistor → Diode (RB) → LED (FB) → Ground

I = 0 A (diode should be OFF)
Vr = 0 V
VLED = 0 V
Vdiode = Vs

***LED is ON in measurement but should be OFF theoretically. This means that the standard diode may have a breakdown voltage at around 7.6V as observed in the experimental data.**

Circuit (e): Vs → Resistor → Zener (FB) → LED (FB) → Ground

Zener behaves similarly to a normal diode in forward bias.
Vr = Vs - 0.7 - 2.0 = Vs - 2.7
Minimum operating voltage is 2.7 V.
I = Vr / 1000

Vr at Vs= 3V, Vr = 3 V - 2.7 V = 0.3V
I = Vr / 1000 = 0.3V/1000 = 0.3 mA
LED turns on, low intensity.

Vr at Vs= 9V, Vr = 9 V - 2.7 V = 6.3V
I = Vr / 1000 = 6.3V/1000 = 6.3 mA
LED turns on, high intensity.

***Follows forward-biased series diode behaviour.**

Circuit (f): Vs → Resistor → Zener (RB) → LED (FB) → Ground
Zener is in reverse bias, conducts only at Vs>Vz.
Vz chosen for our group was 5.0V.

For Vs = 1.5V and 3V, Vs < Vz, act as open circuit, so LED is in OFF condition,
I = 0 A → LED OFF
Vr = I x R = 0
VLED = 0

For Vs = 9V, Vs > Vz, so Zener conducts voltage,
VLED = Vz = 5.0V
Using KVL: Vs = Vr + Vz
Vr = Vs - Vz = 9 - 5 = 4 V
I = Vr / R = 4 V / 1000 Ω = 4 mA

***LED is ON in breakdown, VLED = 5V is sufficient to turn it on brightly.**

COMPARISON BETWEEN MEASURED & THEORETICAL RESULTS

Circuit	V_S (V)	V_{LED} (V)		V_{DIODE} (V)		V_{ZENER} (V)		$V_{RESISTOR}$ (V)	
		Experimental	Theoretical	Experimental	Theoretical	Experimental	Theoretical	Experimental	Theoretical
A	3	3.00	3.00					0.00	0.00
	9	9.00	9.00					0.00	0.00
B	3	1.83	2.00					1.17	1.00
	9	1.95	2.00					7.04	7.00
C	3	1.80	2.00	0.524	0.70			0.672	0.30
	9	1.94	2.00	0.583	0.70			6.46	6.30
D	3	0.0295	0.00	1.69	3.00			0.00	0.00
	9	0.0883	0.00	7.62	9.00			0.00	0.00
E	1.5	1.25	0.00			0.248	1.50	0.00	0.00
	3	1.80	2.00			0.524	0.70	0.672	0.30
	9	1.94	2.00			0.583	0.70	6.46	6.30
F	1.5	0.731	0.00			0.769	1.50	0.00	0.00
	3	1.25	0.00			1.75	3.00	0.00	0.00
	9	1.86	2.00			5.18	5.00	1.96	2.00

From the comparison table, we can see that Circuit A, which only has 1 resistor and 1 light-emitting diode which is the same as the basic circuit, has the same values between theoretical and experimental value. This confirms that basic circuit laws like Kirchoff's Voltage Law (KVL) were being followed and our measurement equipment, which is a multimeter, is working correctly.

For the B, C, E circuit , from the observation, the voltage across the LED/Diode remained relatively constant which is around 1.8V-2.0V for the LED and 0.5V-0.7V for the diode even though the source voltage changed significantly from 3V to 9V. This can be explained due to the non-linear nature of diodes. In forward-biased, diodes will have relatively constant voltage drop, to regulate the voltage for other components in the series circuit. We can also observe that the remaining voltage drop after going through LED and diodes is going to the resistor because $V_{RESISTOR}$ changes with the V_S.

From the observation of the D circuit, the diode is in reverse bias (RB), it acts as an open circuit, preventing current flow, and the LED remains off. This is because the diode acts as a switch or equivalent to an open-circuit, blocking the current flow of the circuit.

For the F circuit, the zener diodes are being reverse-biased. At low V_S, insufficient current flows. At V_s=9V, the Zener diode enters its breakdown region and limits the voltage across itself to ~5V because our group set the Zener voltage to 5V. This confirmed that Zener diodes can be used for voltage regulation once their reverse breakdown threshold is met.

CONCLUSION

In conclusion, the series of experiments successfully investigated the voltage characteristics of both linear (resistors) and non-linear (LEDs, diodes, and Zener diodes) circuit components under varying source voltages. The results consistently validated Kirchhoff's Voltage Law (KVL), as the sum of the experimental voltage drops across all components in each series circuit closely approximated the applied source voltage, confirming a foundational principle of circuit analysis. The key findings was the demonstration of diode behaviour as a non-linear component, where it can maintain a relatively constant voltage drop even though the source voltage changed.

Other than that, the Zener diode in circuit F explains more about the characteristics of diodes by turning into a voltage regulator with its nominal 5V breakdown point. Some small differences between experimental and theoretical values are related to real-world factors like component tolerances, and the use of simplified theoretical models compared to the actual slightly current-dependent voltage of a physical diode.

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